2.3 Matroids

The theory of bases of vector spaces (the Vector Spaces subsection) and transcendence bases of field extensions (the Transcendental Field Extensions subsection) have some formal similarities, as noted in [Wae91]. Here we use the theory of Matroids to formalise this precisely so that the proofs need not be repeated in each case.

Definition 2.3.1 (Matroid)
Consider a set X together with a closure operator $c : \mathcal{P}(X) \to \mathcal{P}(X)$ ("span" operator) such that $c(\varnothing) = \varnothing$. We say

- *$S \subseteq X$ is **independent** if $x \in S \implies x \notin c(S \setminus \{x\})$*
- *$\Gamma \subseteq X$ is **spanning** if $c(\Gamma) = X$.*

Note by definition X is spanning and $\varnothing$ is independent. Moreover all singletons $\{x\}$ are independent.

*We call the pair (X, c) a **matroid** if it also satisfies the following properties*

- ***Finitary** $x \in c(\Gamma) \implies x \in c(\Gamma')$ for some finite subset Γ' of Γ.*
- ***Exchange Property** For all $x, y \in X$ and $Y \subseteq X$ we have*

$$x \in c(Y \cup \{y\}) \setminus c(Y) \implies y \in c(Y \cup \{x\})$$

*We say (X, c) has **finite rank** if it has a finite spanning set.*

*Finally we say $\mathcal{B} \subseteq X$ is a **basis** if it is both **spanning** and **independent**.*

We begin with some elementary characterizations of independent sets

Lemma 2.3.2
Suppose $S \subseteq X$ is a subset

a) $A \subseteq c(S) \implies c(S \cup A) = c(S)$
b) S is independent if and only if no proper subset has the same span.

Proof. We prove each in turn

a) By monotonicity

$$c(S) \subseteq c(S \cup A) \subseteq c(c(S) \cup A) = c(c(S)) = c(S)$$

b) Suppose S is independent and $S' \subsetneq S$ is a proper subset such that $c(S') = c(S)$. Choose $x \in S \setminus S'$ then by definition $x \in S \implies x \in c(S) = c(S') \subseteq c(S \setminus \{x\})$ contradicting independence.

Conversely suppose for some $x \in S$ we have $x \in c(S \setminus \{x\})$. Define $S' \coloneqq S \setminus \{x\}$. Then $x \in c(S')$ implies $c(S) = c(S')$ by a). As S' is a proper subset this contradicts the hypothesis.

■

Lemma 2.3.3
Every subset of an independent set is independent. Furthermore the family of independent sets has finite character.

Proof. The first statement is straightforward. Suppose S a dependent set such that every finite subset is independent. Then there exists $x \in S$ such that $x \in c(S \setminus \{x\})$. Then by the finitary property there exists a finite subset $S' \subseteq S \setminus \{x\}$ such that $x \in c(S')$. Therefore by definition $S' \cup \{x\}$ is not independent, a contradiction. ■

The finitary condition ensures that $\mathcal{E}$ is "inductively ordered"

Corollary 2.3.4
Let $\{S_i\}_{i \in I}$ be a chain of independent subsets. Then $S = \bigcup_{i \in I} S_i$ is also independent.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Let $F \subseteq S$ be a finite subset. For each $x \in F$ choose $i_x \in I$ with $x \in S_{i_x}$. As $\{S_i\}_{i \in I}$ is a chain and F is finite, the sets $\left\{S_{i_x}\right\}_{x \in F}$ are totally ordered by inclusion and so admit a maximum S_{i_0}, whence $F \subseteq S_{i_0}$. By (2.3.3) F is independent, being a subset of the independent set S_{i_0}. As $F \subseteq S$ was an arbitrary finite subset, S is independent by (2.3.3). ■

Lemma 2.3.5 (Extension Property)
Suppose S is an independent set and $x \notin c(S)$, then $S \cup \{x\}$ is independent.

Proof. We require to prove that for all $y \in S$ we have $y \notin c(S \cup \{x\} \setminus \{y\})$. By independence of S we have $y \notin c(S \setminus \{y\})$, so by the Exchange Property $y \in c(S \cup \{x\} \setminus \{y\})$ would imply $x \in c(S)$, contradicting the hypothesis. ■

Proposition 2.3.6
Let $\mathcal{F}$ be a family of independent sets which satisfies the following properties

- $\varnothing \in \mathcal{F}$
- ***extension property*** *$S \in \mathcal{F}$ and $x \notin c(S) \implies S \cup \{x\} \in \mathcal{F}$*
- *$\mathcal{F}$ has finite character*

then $\mathcal{F}$ consists of all independent sets.

Proof. It's enough to show that $\mathcal{F}$ contains all finite independent sets, which follows by induction on $\#S$. For given an independent set S and $x \in S$, then by definition $x \notin c(S \setminus \{x\})$. By the induction hypothesis $S \setminus \{x\} \in \mathcal{F}$ whence by the extension property $S \in \mathcal{F}$ as required. ∎

Proposition 2.3.7 (Basis exists)
Let (X, c) be a matroid, S independent and Γ a subset such that $S \subseteq \Gamma$. Then there exists an independent set $\mathcal{B}$ such that $S \subseteq \mathcal{B} \subseteq \Gamma$ and $c(\mathcal{B}) = c(\Gamma)$.

In particular if Γ is spanning then $\mathcal{B}$ is a basis.

Proof. Consider the collection

$$\mathcal{I} = \{T \text{ independent} \mid S \subseteq T \subseteq \Gamma\}$$

By (2.3.4), $\mathcal{I}$ is chain-complete. Therefore it has a maximal element $\mathcal{B}$ by Zorn's Lemma. Suppose $x \in \Gamma \setminus c(\mathcal{B})$ then $\mathcal{B} \cup \{x\}$ is independent by (2.3.5), contradicting maximality. Therefore $\Gamma \subseteq c(\mathcal{B}) \implies c(\Gamma) \subseteq c(\mathcal{B})$. The reverse inequality is clear so that $c(\Gamma) = c(\mathcal{B})$. ∎

Corollary 2.3.8 (Criteria for bases)
Let (X, c) be a matroid. Then the following are equivalent

a) $\mathcal{B}$ is a basis
b) $\mathcal{B}$ is a minimal spanning set
c) $\mathcal{B}$ is maximally independent (possibly in some spanning set Γ)

Proof. a) $\implies$ b). Let $\mathcal{B}$ be a basis and $\Gamma \subseteq \mathcal{B}$ a spanning set. Then by (2.3.2) b) $\Gamma = \mathcal{B}$.

b) $\implies$ a). Let $\mathcal{B}$ be a minimal spanning set, then by (2.3.7) there exists a subset $\mathcal{B}'$ which is a basis, and in particular spanning. By minimality $\mathcal{B} = \mathcal{B}'$.

c) $\implies$ a). By (2.3.7) there exists a basis $\mathcal{B}'$ containing $\mathcal{B}$. By maximality $\mathcal{B}' = \mathcal{B}$.

a) $\implies$ c). Suppose $\mathcal{B} \subseteq S$ where S is independent. Then S is spanning too, and so by (2.3.2) b) $\mathcal{B} = S$. ∎

Definition 2.3.9 (Sub-matroid)
*A subset $Y \subseteq X$ is a **sub-matroid** if $c(Y) = Y$. In this case $S \subseteq Y \implies c(S) \subseteq Y$ and so we have an induced matroid structure (Y, c).*

Lemma 2.3.10 (Mini Exchange Lemma)
Let $S \subseteq \Gamma$ be finite sets and $x \in X \setminus S$ such that

- *S is independent*
- *$x \in c(\Gamma) \setminus c(S)$*

Then there exists $y \in \Gamma \setminus S$ such that $c(\Gamma \setminus \{y\} \cup \{x\}) = c(\Gamma)$.

Proof. We may assume without loss of generality that $x \notin \Gamma$.

Consider $\tilde{\Gamma} \subseteq \Gamma$ minimal subject to $S \subseteq \tilde{\Gamma}$ and $x \in c(\tilde{\Gamma})$. If $S = \tilde{\Gamma}$ then $x \in c(S)$ a contradiction. Therefore we may choose $y \in \tilde{\Gamma} \setminus S$. By minimality we have $x \in c(\tilde{\Gamma}) \setminus c(\tilde{\Gamma} \setminus \{y\})$. Therefore by the Exchange Property we have $y \in c(\tilde{\Gamma} \setminus \{y\} \cup \{x\})$. Then by (2.3.2) applied twice

$$c(\Gamma \setminus \{y\} \cup \{x\}) = c(\Gamma \cup \{x\}) = c(\Gamma)$$

as required. ∎

Proposition 2.3.11 (Exchange Lemma)
Let S be an independent set and Γ be a finite set such that $S \subseteq c(\Gamma)$. Then there exists a subset $T \subseteq \Gamma$ such that

- *$\#T = \#S$*

- $c(\Gamma \setminus T \cup S) = c(\Gamma)$.

In particular $\#S \leq \#\Gamma$.

Proof. By considering the sub-matroid $(c(\Gamma), c)$ we may assume without loss of generality that $X = c(\Gamma)$.

Let $n = \#\Gamma$ and consider the set of pairs

$$\mathcal{F} \coloneqq \{(A, B) \mid A \subseteq S, \quad B \subseteq \Gamma, \quad \#A = \#B, \quad c(\Gamma \setminus B \cup A) = X\}$$

Essentially $\mathcal{F}$ is the set of swaps we may perform from S to Γ whilst preserving the span. It is non-empty because $(\varnothing, \varnothing) \in \mathcal{F}$ and we wish to show that $(S, B) \in \mathcal{F}$ for some B.

Observe that for all $(A, B) \in \mathcal{F}$ we have $\#B \leq n$ so choose a pair such that $\#B$ is maximal. We wish to show that in this case $A = S$, so we suppose to the contrary that $A \subsetneq S$.

We claim that $B \subsetneq \Gamma$, for $B = \Gamma$ implies by construction $c(A) = X$ which would imply $c(A) = c(S)$, contradicting the criteria for independence of S given by (2.3.2) b).

Define $\Gamma' \coloneqq \Gamma \setminus B \cup A$. Then by assumption $c(\Gamma') = X$ and $A \subseteq \Gamma'$ is independent. Choose $x \in S \setminus A$ then by definition of independence $x \notin c(A)$. By (2.3.10) there exists $y \in \Gamma' \setminus A$ such that

$$c(\Gamma' \setminus \{y\} \cup \{x\}) = c(\Gamma') = X$$

Note by construction that $y \notin B$, so we see that $(A \cup \{x\}, B \cup \{y\}) \in \mathcal{F}$ has greater length, which contradicts maximality. ∎

Corollary 2.3.12 (Bases have the same cardinality)
Every base of a finite rank matroid is finite and of the same size. Denote this by $r(X)$.

Proof. There is a finite basis by (2.3.7). Then apply (2.3.11). ∎

Corollary 2.3.13
Let (X, c) *be a finite-rank matroid and* $S \subseteq X$ *an independent subset, then* $\#S \leq r(X)$. *Similarly a spanning subset* Γ *satisfies* $\#\Gamma \geq r(X)$.

Proof. Apply (2.3.7) and (2.3.12). ∎

Corollary 2.3.14 (Basis Criteria)
Let $\mathcal{B}$ *be a subset of a finite-rank matroid* (X, c). *Then the following are equivalent*

a) $\mathcal{B}$ *is a basis*
b) $\mathcal{B}$ *is independent and* $\#\mathcal{B} \geq r(X)$
c) $\mathcal{B}$ *is spanning and* $\#\mathcal{B} \leq r(X)$

Proof. Apply (2.3.7) and (2.3.12). ∎

Proposition 2.3.15
Let $Y \subseteq X$ *be a sub-matroid of a finite-rank matroid. Then* $Y = X \iff r(Y) = r(X)$.

Proof. Let $\mathcal{B}$ be a basis for Y then $\#\mathcal{B} = r(Y) = r(X)$ and is a-fortiori independent in X. Therefore by (2.3.14) $\mathcal{B}$ is a basis for X and hence $X = c(\mathcal{B}) = Y$. ∎